

- ## Newton's Laws

Newton's laws are a lot to take in. We should establish a common nomenclature before continuing. As a group, define each of the following and provide an example where appropriate. Check your answers with your instructor before moving on:

- **Inertia:** the propensity of an object to move.
- **Inertial Reference Frame:** the reference frame in which a body/object experiences sum zero forces or no acceleration.
- **Force:** the interaction required to make an object move (accelerate).
- **Normal Force:** The stabilizing force exerted on an object preventing the object from changing position with respect to the direction of application (generally perpendicular to the direction of motion).
- **Weight (force due to gravity):** Force applied to objects close to the surface of a large body (earth).
- **Tension:** Pulling force, induced by the object, in the direction of application (cannot be a pushing force).
- **Dry Friction:** Resistance force applied in the opposite direction of an object's motion requiring surface contact.
- **Resistance/Drag:** Contact force on an object traveling through a liquid dependent on velocity and surface area and applied in the direction opposite of the object's motion.
- **Bouyancy:** Contact force applied to a body dependent on viscosity and surface area, much like the normal force. This force is applied normal to the direction of motion as well.

Force	Abbreviation	Relationship?	Velocity Dep?	Contact?	Diagram
Gravity					
Tension					
Normal					
Spring					
Friction(s)					
Drag					
'Centripetal'					